implementing and analyzing a basic rl algorithm

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1. objective

this report documents the implementation and analysis of a 5x5 gridworld mdp solved using three approaches: linear system solving, value iteration, and policy iteration.

2. environment

2.1 environment parameters

parameter	value
grid size	5x5 (25 cells total)
goal state	cell 14 (reward +5.0)
danger states	cells 2, 18, 21 (reward -5.0)
blocked cells	cells 6, 7, 11, 12
valid states	17
actions	up, right, down, left

2.2 mdp components

state space: cells 0-24 (row-major), excluding blocked cells 6,7,11,12.

action space: four moves; hitting wall/blocked = stay in place.

transition model: probability (1-noise) moves intended direction; probability noise moves random.

reward function: goal +5.0 (terminal), danger -5.0 (terminal), others 0.0.

3. implementation

3.1 gridworld class

the `gridworld` class implements: `__init__`, `_state_from_action`, `is_terminal`, `get_states`, `get_actions`, `get_reward`, `get_transitions`, `solve_linear_system`.

3.2 key details

- state indexing: row-major
- noise=0.0: deterministic; noise=0.2: 0.8 intended, 0.067 each other

4. methods

4.1 linear system solver

$v = r + \gamma \times p \times v \rightarrow ax = b$ where $a = i - \gamma p/4$, b = expected rewards. solved via `np.linalg.solve()`.

4.2 value iteration

$v(s) \leftarrow \max_a \Sigma\, p(s'|s,a)[r(s') + \gamma v(s')]$. iterate until delta < tolerance.

4.3 policy iteration

alternates: policy evaluation (v = r + γpv) and policy improvement (π(s) = argmax_a q(s,a)).

5. results

5.1 linear system solver

γ = 0.95:

```
 -2.69  -3.60  -5.00  -1.64  -0.00
-2.35   ----   ----  -0.00   1.64
-2.51   ----   ----  -0.00   goal
-3.19  -3.91  -4.10  -5.00  -0.57
-3.82  -5.00  -3.99  -3.45  -1.82
```

γ = 0.75:

```
 -0.78  -2.24  -5.00  -1.54  -0.00
-0.38   ----   ----  -0.00   1.54
-0.48   ----   ----  -0.00   goal
-1.22  -2.45  -2.73  -5.00  -0.17
-2.37  -5.00  -2.71  -2.34  -0.75
```

5.2 value iteration

configuration	iterations
γ=0.95, noise=0.0	12
γ=0.75, noise=0.0	12
γ=0.95, noise=0.2	15
γ=0.75, noise=0.2	7

γ=0.95, noise=0.0:

```
  3.15   2.99  -5.00   4.51   4.75
 3.32   ----   ----   4.75   5.00
 3.49   ----   ----   5.00   goal
 3.68   3.87   4.07  -5.00   5.00
 3.49  -5.00   4.29   4.51   4.75
```

value iteration γ=0.95 noise=0.0

γ=0.95, noise=0.2:

```
  0.42  -0.10  -5.00   3.60   4.51
 0.56   ----   ----   4.49   4.88
 0.65   ----   ----   4.22   goal
 0.72   0.83   1.40  -5.00   4.17
 0.23  -5.00   2.09   2.90   3.84
```

value iteration γ=0.95 noise=0.2

5.3 policy iteration

γ=0.95, noise=0.0 (10 iterations):

```
 3.15   2.99  -5.00   4.51   4.75
 3.32   ----   ----   4.75   5.00
 3.49   ----   ----   5.00   goal
 3.68   3.87   4.07  -5.00   5.00
 3.49  -5.00   4.29   4.51   4.75
```

policy iteration   optimal policy

5.4 optimal policy map

state	action	rationale
0	down	toward goal
1	left	avoid blocked
3,8,13	right	approach goal
4,9,19,24	down/up	reach goal

6. convergence analysis

algorithm	config	iterations
linear solver	γ=0.95	1
linear solver	γ=0.75	1
value iteration	γ=0.95, noise=0.0	12
value iteration	γ=0.75, noise=0.0	12
value iteration	γ=0.95, noise=0.2	15
value iteration	γ=0.75, noise=0.2	7
policy iteration	γ=0.95, noise=0.0	10

factors affecting convergence: - discount factor γ: higher γ → more iterations - noise: increases uncertainty, more iterations - grid topology: blocked cells affect value propagation

7. algorithm comparison

linear solver: closed-form, deterministic only, random policy baseline.

value iteration: simple, works for stochastic, max over actions each update.

policy iteration: fewer outer iterations, explicit policy at each step.

all three produce correct results; vi and pi reach identical optimal policy.

linear solver γ=0.95

references: sutton & barto (2018) reinforcement learning: an introduction - https://web.stanford.edu/class/psych209/Readings/SuttonBartoIPRLBook2ndEd.pdf